



MTU Integrated Automation Module

MTU-Module

Controller

1. PCD 2-01	B00E51000005	ILC 150 ETH	Programmable Control Device
2. PCD 3-01	B00E51000006	ILC 170 ETH	Programmable Control Device
3. PCD 4-01	B00E51000007	ILC 370 PN 2 TX-IB/M	Programmable Control Device

Power Supply

1. PID 1-01	B00E51000025	IB IL 24 PWR IN-PAC	Power Input Device
2. PID 2-01	B00E51000026	IB IL 24 PWR IN/RIL-0,8-PAC	Power Input Device

Bus Coupler

1. BID 1-01	B00E51000008	IBS IL 24 RB-T-PAC	Bus Interface Device
2. BID 2-01	B00E51000009	IBS IL 24 BK-T/U-PAC	Bus Interface Device
3. BID 3-01	B00E51000010	IL PN BK DI8 DO4 2TX-PAC	Bus Interface Device

I/O Analog

1. AID 1-01	B00E51000016	IB IL AI 2/SF-PAC	Analog Input Device
2. AID 2-01	B00E51000017	IB IL AI 8/SF-PAC	Analog Input Device
3. AID 3-01	X00E50210116	IB IL AI 4/EF-PAC	Analog Input Device
4. AOD 1-01	B00E51000021	IB IL AO 2/SF-PAC	Analog Output Device
5. AOD 3-01	B00E51000022	IB IL AO 2/U/BP-PAC	Analog Output Device
6. TID 1-01	B00E51000019	IB IL TEMP 2 RTD-PAC	Temperature Input Device
7. TID 2-01	B00E51000020	IB IL TEMP 4/8 RTD-PAC	Temperature Input Device
8. TID 3-01	B00E51000018	IB IL TEMP 2 UTH-PAC	Temperature Input Device

I/O Digital

1. DID 1-01	B00E51000011	IB IL 24 DI 4-PAC	Digital Input Device
2. DID 2-01	B00E51000012	IB IL 24 DI 16-PAC/SN	Digital Input Device
3. DOD 1-01	B00E51000013	IB IL 24 DO 4-PAC	Digital Output Device
4. DOD 2-01	B00E51000014	IB IL 24 DO 16-PAC/SN	Digital Output Device
5. DOD 3-01	B00E51000015	IB IL 24/230 DOR4/W-PAC	Digital Output Device
6. DOD 4-01	B00E51000050	IB IL 24 DO 8-NPN-PAC	Digital Output Device

Functional I/O, Interfaces

1. FID 1-01 _{CNT}	B00E51000023	IB IL CNT-PAC	Frequency Onput Device
2. PWD 1-01 _{PWM}	B00E51000024	IB IL PWM/2-PAC	Pulse Wide Device
3. SID 1-01 _{RS232}	B00E51000044	IB IL RS 232-PRO-PAC	Serial Interface Device
4. SID 2-01 _{RS485-422}	B00E51000045	IB IL RS 485/422-PRO-PAC	Serial Interface Device
5. SID 3-01 _{CAN-Master}	B00E51000046	IB IL CAN-MA-PAC	CAN-Interface



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Colour "Diagnostic Area"	Group	Modules MTU designation
gray	Bus Coupler	BID
black	Power Supply	PID
green	Analog Input	AID, TID
blue	Digital Input	DID
yellow	Analog Output	AOD
pink	Digital Output	DOD
brown/orange	Functional I/O, Interfaces	SID, FID, PWD



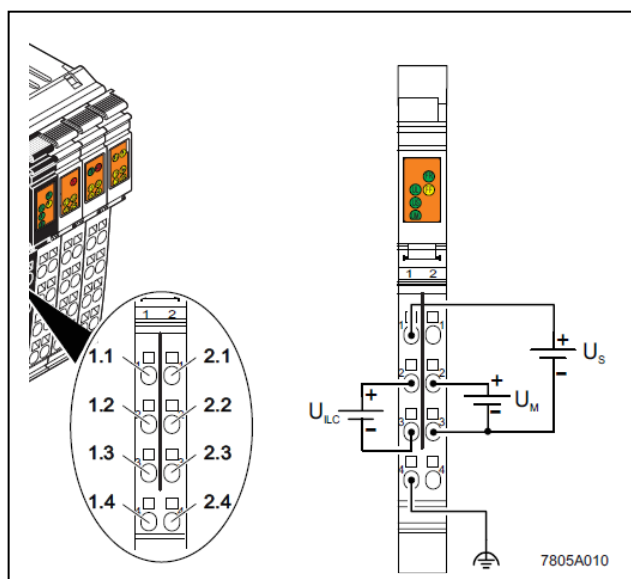


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Controller

Name	PCD 2-01	PCD 3-01	PCD 4-01
Device			
Part No.	B00E51000005	B00E51000006	B00E51000007
Memory	-	SD-Card / 256MB	CF-Card / 256MB
Interfaces	- 1x ETH 10/100 - 1x PRG (RS-232)	- 2x ETH 10/100 - 1x PRG (RS-232)	- 2x ETH 10/100 - 1x PRG (RS-232) - 2x Interbus
Application	Duty Alarm System „DAS“ (max. 8 Station)	- Low Cost (Marine) - Comm.Redundancy - Interface Modbus	- High End (Marine) - Hot Standby Red. - DAS (max.16 Station)

Power supply connection PCD 2-01, PCD 3-01 and PCD 4-01



- Supply the Inline controller using external 24 V DC voltage sources. The permissible voltage ranges from 19.2 V DC to 30 V DC (ripple included). The power consumption of the Controller at 24 V is typically 4.8 W (no local bus devices connected).
1. Connect the power supplies to the connector for power supply as shown in Figure 2-22.
 2. Insert the connector in the Inline controller.
 3. Switch on the power supplies.
 4. The U_L , U_M , and U_S LEDs light up and, after around 10 seconds, the FR and RDY LEDs start flashing.

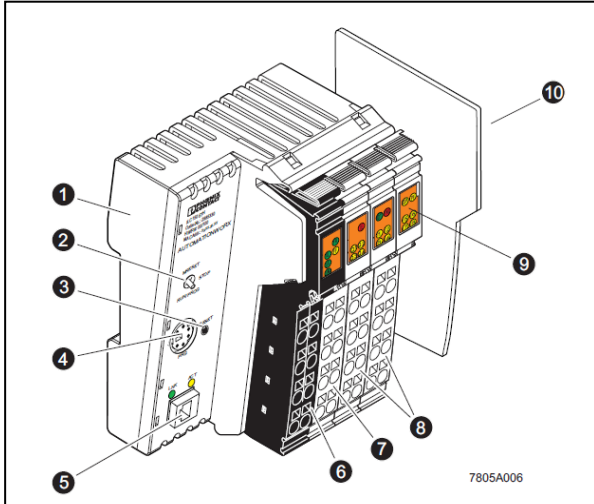
The Inline controller is now fully initialized.
If the LEDs do not light up or start flashing, there is a serious fault on the Inline controller.





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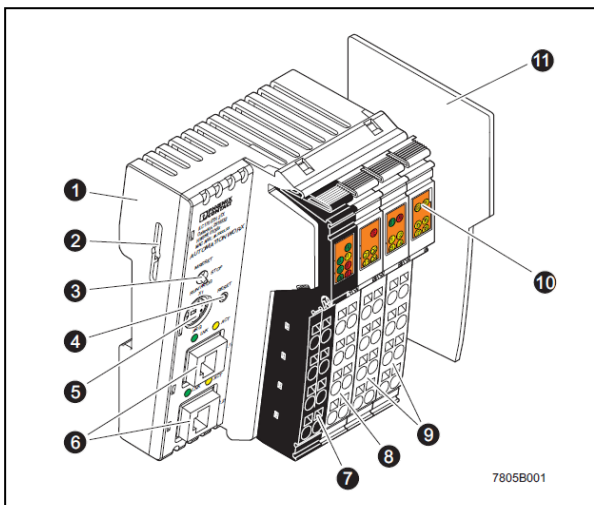
Connection and operating elements PCD 2-01



The Inline controller consists of the following components:

1. Electronics base
2. Mode selector switch
3. Reset button
4. V.24 (RS-232) interface
5. Ethernet connection
6. Connector 1: terminal points for voltage supply
7. Connector 2: output terminal points
8. Connectors 3 and 4: input terminal points
9. Diagnostic and status indicators
10. End plate

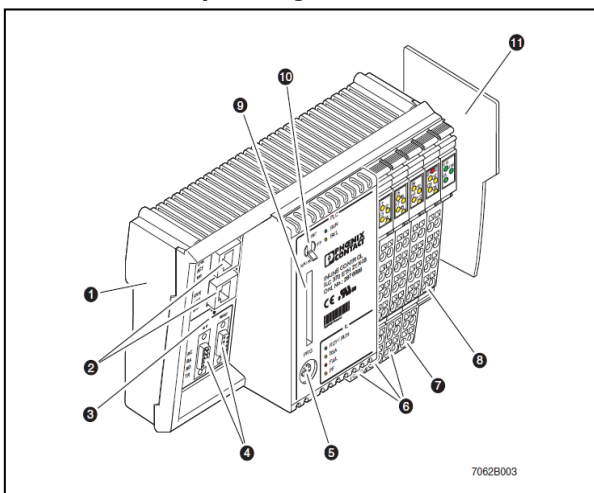
Connection and operating elements PCD 3-01



The Inline controller consists of the following components:

1. Electronics base
2. Slot for the parameterization memory/card holder (SD card).
3. Mode selector switch
4. Reset button
5. V.24 (RS-232) interface (X1)
6. Ethernet interfaces (X2.1/X2.2)
7. Connector 1: terminal points for voltage supply
8. Connector 2: output terminal points
9. Connectors 3 and 4: input terminal points
10. Diagnostic and status indicators
11. End plate

Connection and operating elements PCD 4-01



The Inline controller consists of the following components:

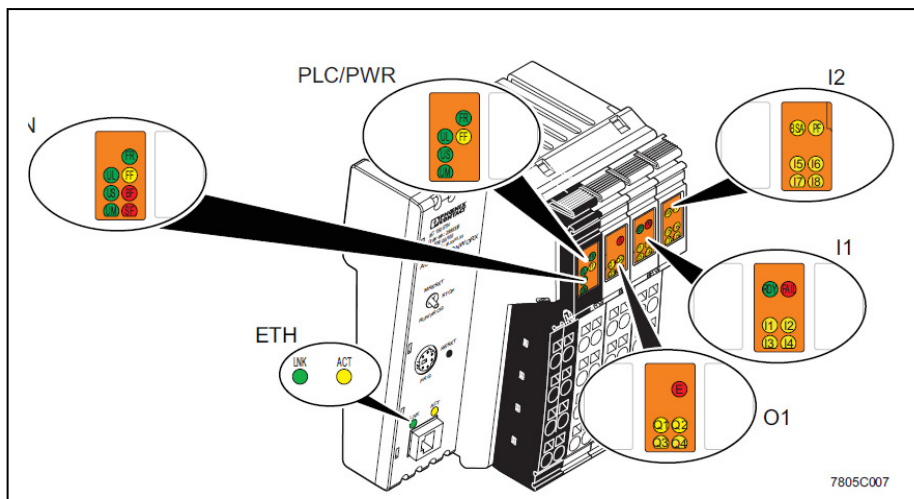
1. Electronics base
2. Ethernet connection
3. Reset button
4. INTERBUS connection (higher-level INTERBUS)
5. RS-232 interface
6. Connectors 1 to 3: input terminal points
7. Connector 4: output terminal points
8. Connector 5: terminal points for voltage supply
9. Card holder for parameterization memory
10. Mode selector switch
11. End plate





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Diagnostic and status indicators PCD 3-01



Des.	Color	Status	Meaning
SF	Red		System error message (SystemFail)
		OFF	System error message/PROFINET diagnostics not present.
		ON	System error message/PROFINET diagnostics present.
PLC: Diagnostics of the Inline controller			
FR	Green		Inline controller running
		OFF	IEC 61131 runtime system not ready to operate.
		Flashing	IEC 61131 runtime system successfully initialized. Control function in READY/STOP state, program not processed.
		ON	IEC 61131 runtime system successfully initialized and a program is running. Control function in RUN state.
FF	Yellow		Failure
		ON	A runtime error has occurred in the IEC 61131 runtime system program
		OFF	No runtime error has occurred in the IEC 61131 runtime system program
FR + FF		Flashing	Applies to ILC 170 ETH 2TX and ILC 190 ETH 2TX only: There is no SD card in the Inline controller.
PWR: Supply voltage (See also Section "Remote bus" on page 2-25)			
UL	Green		24 V supply U_{ILC} for generation of the voltages U_L and U_{ANA}
		OFF	Supply voltage not present
		ON	Supply voltage present (indication if 24 V supply voltage U_{ILC} present)
US	Green		24 V supply for segment circuit
		OFF	Supply voltage not present
		ON	Supply voltage present
UM	Green		24 V supply for main circuit
		OFF	Supply voltage not present
		ON	Supply voltage present
IL: INTERBUS diagnostics			
RDY	Green		INTERBUS master ready to operate/data transmission active (INTERBUS ready/running)
		OFF	INTERBUS master not ready to operate.
		Flashing	INTERBUS master in READY or ACTIVE state.
		ON	INTERBUS master in RUN state.
FAIL	Red		Failure
		OFF	No error occurred
		ON	One of the following errors has occurred: – Bus error in the connected bus (remote bus/local bus) – Controller error



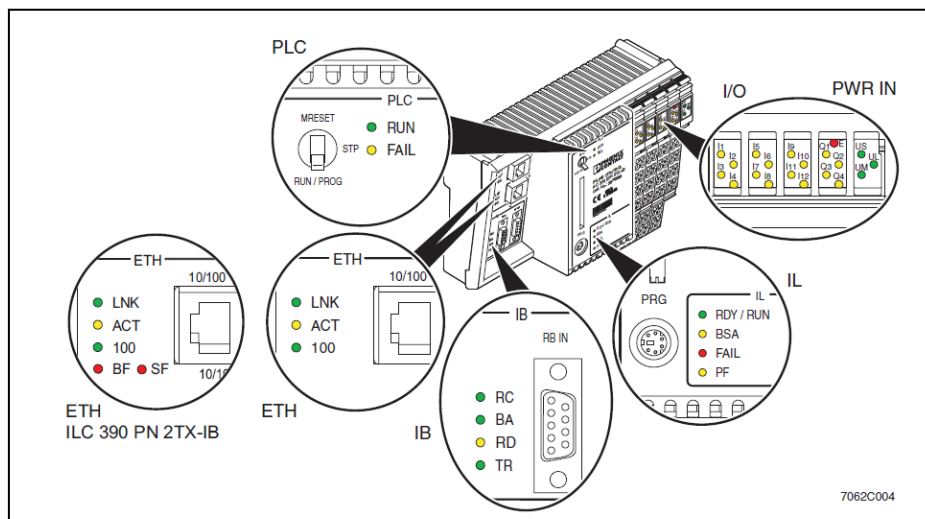


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Des.	Color	Status	Meaning
BSA	Yellow		Bus segment aborted
		OFF	Bus segment(s) in the connected bus not switched off.
		ON	One or more bus segments in the connected bus are switched off.
PF	Yellow		Peripheral fault
		OFF	Peripheral fault on a device in the connected bus
		ON	Peripheral fault of a device in the connected bus (local bus or remote bus)
I/O: Digital inputs and outputs			
I1 to I8	Yellow		Inputs 1 to 8
		OFF	Corresponding input not set
		ON	Corresponding input is set
E	Yellow		Error
		OFF	No short circuit/overload at one of the outputs 1 to 4
		ON	Short circuit/overload at one of the outputs 1 to 4
Q1 to Q4	Yellow		Outputs 1 to 4
		OFF	Corresponding output not set
		ON	Corresponding output is set

Des.	Color	Status	Meaning
ETH: State of the Ethernet interface (applies to both interfaces (X2.1, X2.2) for the ILC 170 ETH 2TX/ILC 190 ETH 2TX)			
LNK	Green	OFF	Connection not established successfully
		ON	Connection established successfully (link): The Inline controller is able to contact another network device.
ACT	Yellow	OFF	Data transmission inactive
		ON	Data transmission (activity): The Ethernet interface transmits or receives data.
PROFINET: (ILC 170 ETH 2TX/ILC 190 ETH 2TX)			
BF	Red		Status of PROFINET IO communication/communication error (BusFail)
		OFF	The PROFINET IO controller has established an active communication connection to at least one PROFINET IO device.
		ON	No PROFINET IO communication available (no link status at the Ethernet ports).
		Flashing	No communication connection to the PROFINET IO controller, the connection is being established. Link status available.

Diagnostic and status indicators PCD 4-01





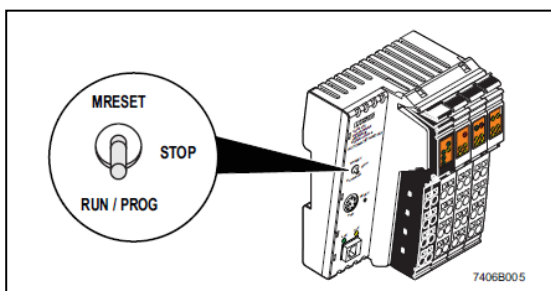
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Des.	Color	Meaning
ETH: State of the Ethernet Interface		
LNK	Green	Connection established successfully (link)
	ON:	The Inline Controller is able to contact another network device.
ACT	Yellow	Data transmission (activity)
	ON:	Ethernet interface transmits or receives data
100	Green	Transmission speed
	ON:	100 Mbps
	OFF:	10 Mbps (if LNK is ON)

Des.	Color	Meaning
Only ILC 390 PN 2TX-IB:		
 The firmware version 1.40 does not support the function of these LEDs. This means that the LEDs are off for a controller with this firmware version.		
BF	Red	Communication error
	ON:	No link status available.
	Flashing:	Link status available, no communication connection to the I/O controller, the connection establishment is active right now.
	OFF:	The I/O controller established an active communication connection to the I/O device.
SF	Red	System error
	ON:	System error occurred (faulty parameterization, bus error, I/O error)
	OFF:	No system error, INTERBUS operating without errors
PLC: Diagnostics of the Inline Controller		
RUN	Green	Inline Controller running
	OFF:	IEC 61131 runtime system not ready to operate.
	Flashing:	IEC 61131 runtime system successfully initialized. Control function in READY/STOP state, program not processed.
	ON:	IEC 61131 runtime system successfully initialized and a program is running. Control function in RUN state.
FAIL	Yellow	Failure
	ON:	A runtime error occurred in the IEC 61131 runtime system program.
	OFF:	No runtime error occurred in the IEC 61131 runtime system program.

IB: Diagnostics of the Higher-Level INTERBUS System (INTERBUS Slave of the Inline Controller)		
RC	Green	Remote bus connection (remote bus check)
	ON:	Higher-level controller board is connected.
	OFF:	Connection to the higher-level controller board interrupted or not yet established.
BA	Green	Remote bus active
	ON:	Data transmission on INTERBUS active (status of the higher-level controller board: RUN)
	OFF:	No data transmission
RD	Yellow	Remote bus disabled
	ON:	The outgoing remote bus interface is disabled.
	OFF:	The outgoing remote bus interface is not disabled.
TR	Green	PCP active (transmission) (currently without function)
	ON:	Inline Controller receiving or sending PCP data
	OFF:	Inline Controller not receiving or sending PCP data

Mode selector switch



Operating mode	Explanation
RUN/PROG	The application program is in the RUN state. The PC WorX/PC WorX Express software can be used for program and configuration modifications. The monitoring and online functions can be used.
STOP	The application program is in the STOP state.
MRESET	Retain data and the application program are deleted. Set the mode selector switch in the following sequence to perform deletion of retain data and the application program: - Set the switch to the MRESET position for three seconds. - Release the switch for less than three seconds. - Set the switch to the MRESET position for three seconds.





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Power Input Device

Name	PID 1-01	PID 2-01		
Device				
Part No.	B00E51000025	B00E51000026		
Channel	1	1		
Signals / Sensors	- Supply U_M (24V) (Main Circuit) - Supply U_S (24V) (Segment Circuit)	- Supply U_L (Logic Circuit / 24V $\rightarrow$ 7,5V)		

1. PID 1-01

Features

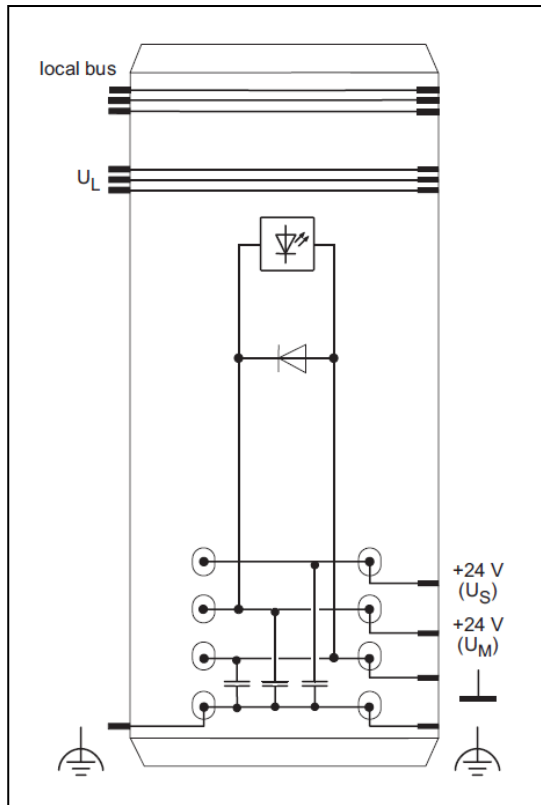
- Supply of the 24 V main voltage U_M
- Supply/provision of the 24 V segment voltage U_S
- Main circuit protected by an external fuse
- Segment circuit can be protected by an external fuse
- Diagnostic indicator
- Approved as power terminal following a safety segment circuit



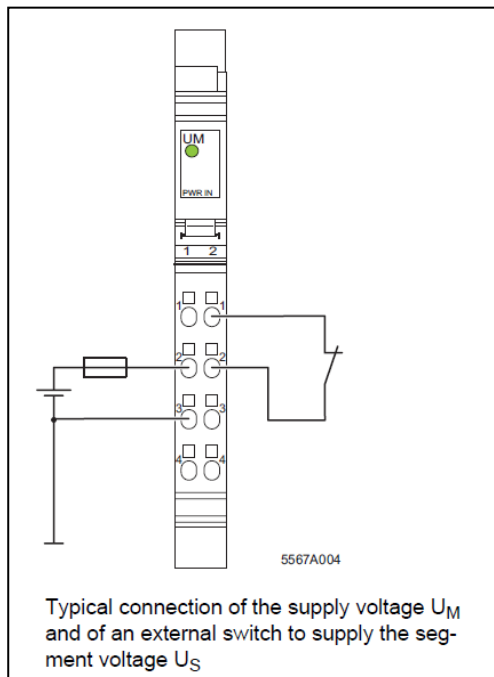


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1.1 Internal circuit diagram PID 1-01



1.2 Connection example



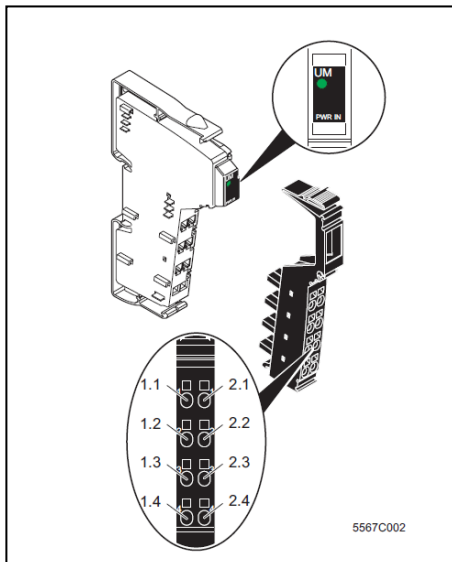
- Protect the 24 V supply with an external fuse!
- To ensure maximum current carrying capacity, use a power connector to connect the cables. In these connectors the adjacent terminal points 1.2 and 2.2, 1.3 and 2.3 are jumpered internally.
- Most I/O terminals receive their supply voltage from the segment circuit.
- The switch can be used to create a switched segment circuit.
- If this is not needed for your application, you can provide the segment voltage in one of the following ways:
 - Jumper connections 1.1 and 1.2 or 2.1 and 2.2
 - Supply the segment voltage separately
 - Use an additional segment terminal





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1.2 Diagnostic/status indicators and terminal point assignment



Local diagnostic indicators

Des.	Color	Meaning
UM	Green	24 V Voltage (main circuit U_M)

Terminal point assignment

Terminal Point	Assignment
1.1, 2.1	Supply points for the segment circuit U_S (+24 V) Connection of a switch or a jumper in the segmentation level.
1.2, 2.2	Supply points for the main circuit U_M (Main Circuit; +24 V) Connection of a switch or a jumper in the segmentation level. These terminal points are connected with each other and with the potential jumper of the unprotected main supply U_M . The potential jumpers of the unprotected main circuit U_M and the segment circuit U_S have a combined current carrying capacity of 8 A.
1.3, 2.3	Ground contact (GND) The reference potential is directly led to the potential jumper and is, at the same time, ground reference for the main and segment voltage.
1.4, 2.4	FE connection The contacts are directly connected with the potential jumper and the FE spring on the bottom of the housing. The terminal is grounded when it is snapped onto a grounded DIN rail.
	Terminal points 1.1, 1.2 and 1.3 are connected with a capacitor to FE.